

Exploratory data analysis for an online store

The study is based on a dataset containing information about product sales across various traffic channels, browsers, countries, continents, and user languages.

The goal of the analysis is to identify key trends in sales dynamics, determine product categories that contribute most to the company's total revenue, and explore relationships between metrics across channels, categories, and continents, with an assessment of their statistical significance.

Detailed calculations are available at:

 [Google Colab](#)

The analytical dashboard is published on Tableau Public:

 [Tableau Dashboard](#)

Dataset Description

- Total rows: 349,545
- Total columns: 18
- Numeric columns (5): `ga_session_id`, `account_id`, `is_verified`, `is_unsubscribed`, `price`
- Categorical columns (12): `continent`, `country`, `device`, `browser`, `mobile_model_name`, `operating_system`, `language`, `channel`, `medium`, `category`, `name`, `short_description`
- Datetime columns (1): `event_date`
- Unique sessions: 349,545
- Time period covered: from 2020-11-01 to 2021-01-31

Missing Values

- `language`: 114,266 missing
- `account_id`, `is_verified`, `is_unsubscribed`: 321,600 missing each
- `category`, `name`, `price`, `short_description`: 316,007 missing each
- Most missing values relate to user registration data, since the dataset includes all sessions (349,545), but only 27,945 sessions involved user registration — meaning registration data is present in only 27,945 rows.
- Additionally, the number of sessions with completed orders is 33,538.
- No duplicates were found.

General Trends

- Total sales during the period: \$31,971,731.10
- Average order value: \$953
- Minimum order value: \$3
- Maximum order value: \$9,585
- Total number of orders: 33,538
- Most orders (17,492 or 52.0%) fall within the price range of \$0 to \$479.25

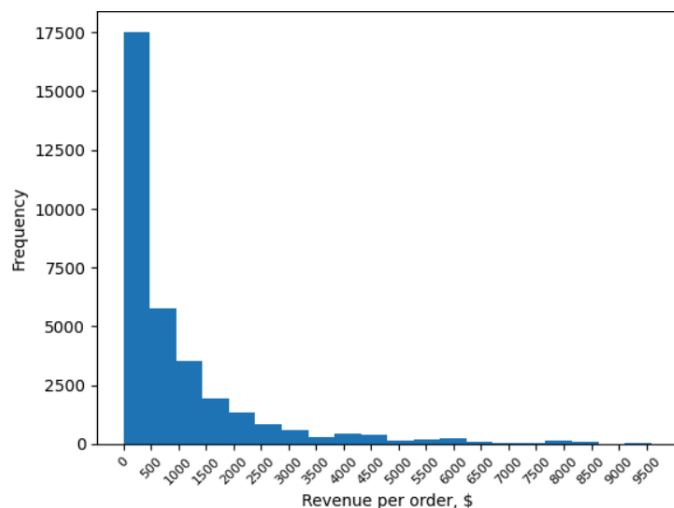


Figure 1. Distribution of Order Size

The highest revenue (\$17,665,280), as well as the largest number of orders (18,553), came from sales to consumers on the continent of America. Overall, the top three continents by revenue are America, Asia, and Europe. The average order value across these continents differs only slightly. Key performance indicators are presented in Figures 2, 3, and 4.

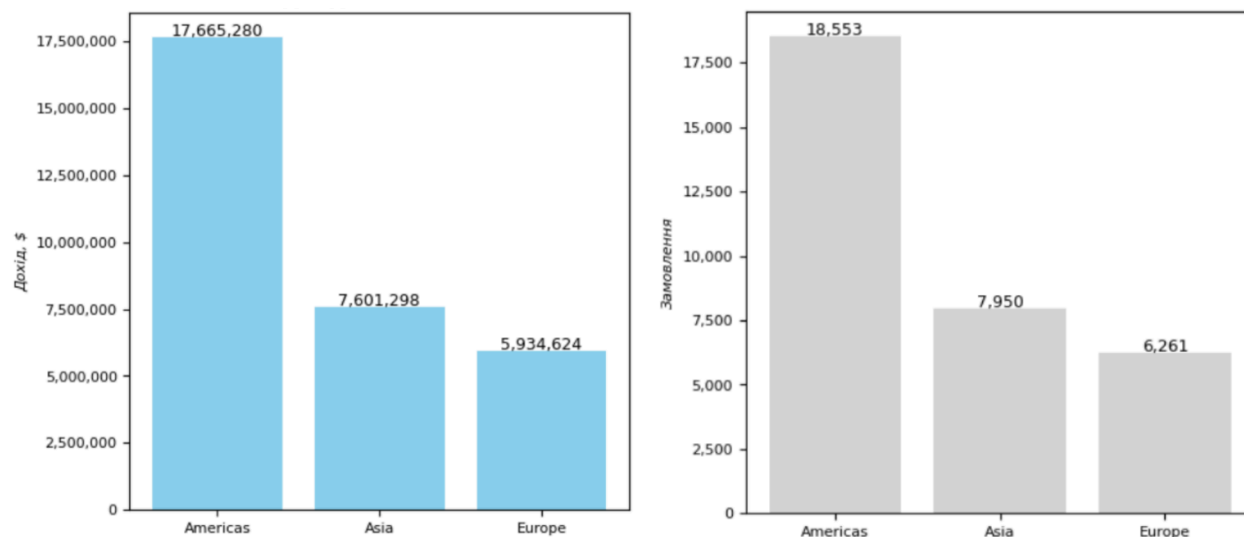


Figure 2. Top 3 Continents by Revenue

Figure 3. Top 3 Continents by Number of Orders

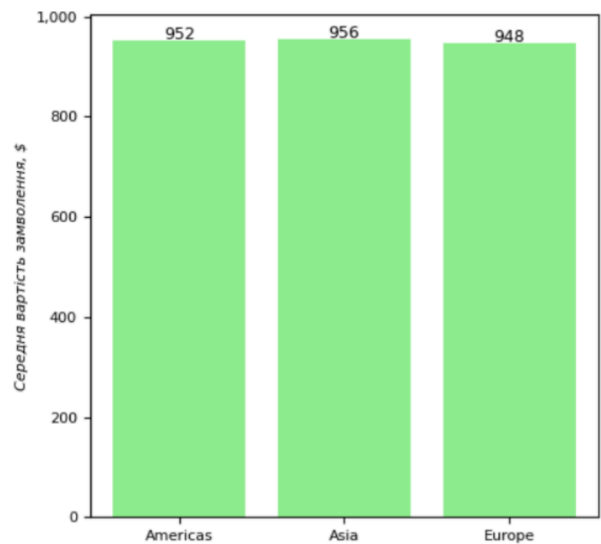


Figure 4. Average Order Value Across the Top 3 Continents

Key sales indicators by country are presented in Figures 5, 6, and 7. The United States ranks first both in terms of revenue (\$13,943,553) and number of orders (14,673). However, the highest average order value was recorded in France — \$1,048.

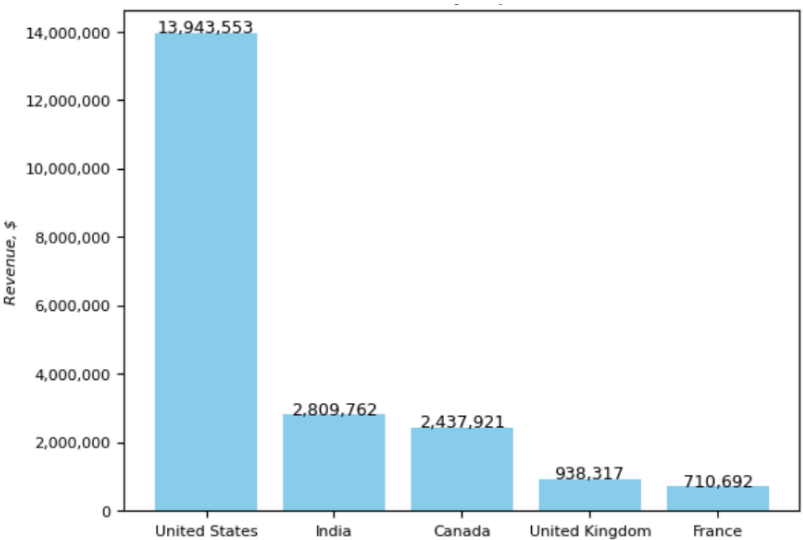


Figure 5. Top 5 Countries by Revenue

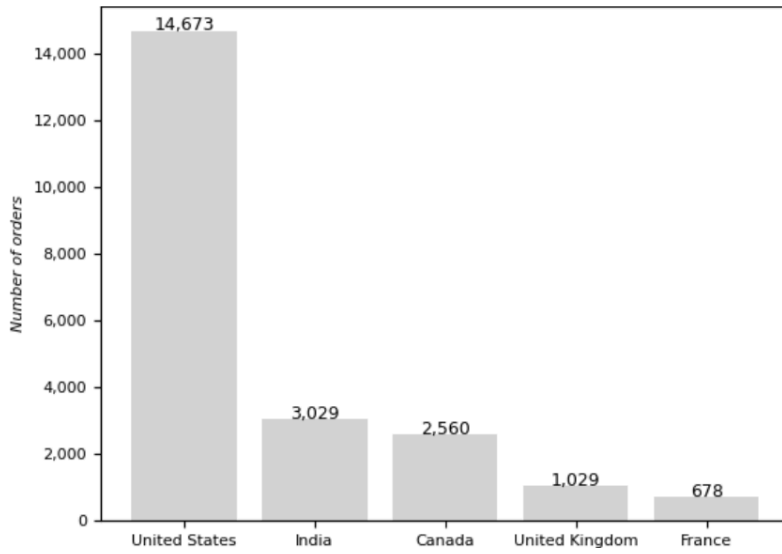


Figure 6. Top 5 Countries by Number of Orders

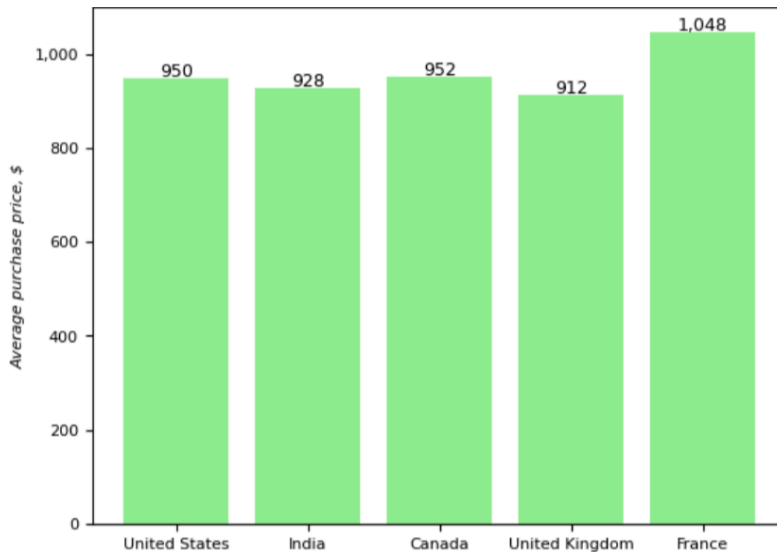


Figure 7. Average Order Value Across the Top 5 Countries

The highest revenue in the United States came from the sale of products in the "Sofa & armchairs" category, totaling \$3,707,144.5 from 1,903 orders.

index	category	revenue	count
1	Sofas & armchairs	3,707,144	1,903
2	Chairs	2,619,774	2,576
3	Beds	2,213,058	1,298
4	Bookcases & shelving units	1,567,607	3,374
5	Cabinets & cupboards	994,546	995
6	Outdoor furniture	929,245	984
7	Tables & desks	777,865	1,248
8	Chests of drawers & drawer units	382,388	616
9	Bar furniture	330,805	487
10	Children's furniture	207,575	752

Table 1. Top 10 Product Categories by Revenue in the Highest-Grossing Country (USA)

The global ranking of product categories by revenue is also led by the "Sofas & armchairs" category, with \$8,388,254.5.

index	category	revenue	count
1	Sofas & armchairs	8,388,254	4,301
2	Chairs	6,147,749	5,952
3	Beds	4,919,725	2,926
4	Bookcases & shelving units	3,640,818	7,630
5	Cabinets & cupboards	2,336,500	2,318
6	Outdoor furniture	2,142,222	2,229
7	Tables & desks	1,790,308	2,941
8	Chests of drawers & drawer units	906,562	1,452
9	Bar furniture	735,503	1,092
10	Children's furniture	467,697	1,702

Table 2. Top 10 Product Categories by Total Sales Revenue

Full details on revenue by category and country are available in Table 3.

country	United States	India	Canada	United Kingdom	France	Total
Sofas & armchairs	3707144.5	788430.0	692427.5	234812.0	187735.0	5610549.0
Chairs	2619773.8	544309.2	417740.8	188519.4	134029.4	3904372.6
Beds	2213058.0	358319.5	354772.0	133816.0	116414.0	3176379.5
Bookcases & shelving units	1567606.9	364507.4	278981.9	113987.6	73830.0	2398913.8
Cabinets & cupboards	994545.5	191888.0	181802.0	71684.5	59101.5	1499021.5
Outdoor furniture	929245.2	162289.4	185322.8	57002.4	40486.4	1374346.2
Tables & desks	777865.0	186157.5	132678.0	49374.0	42299.0	1188373.5
Chests of drawers & drawer units	382388.0	73111.0	71952.0	36784.0	21544.5	585779.5
Bar furniture	330805.0	57657.0	51724.0	22103.0	11199.0	473488.0
Children's furniture	207575.0	39177.0	30264.0	13348.0	14258.0	304622.0
Total	13730006.9	2765846.0	2397665.0	921430.9	700896.8	20515845.6

Table 3. Total Sales by Product Categories (Top 10 Categories) in the Top 5 Countries by Revenue

In terms of devices, the highest revenue (\$18,864,039.0) and the largest number of orders (19,702) came from desktop users (see Figures 8 and 9).

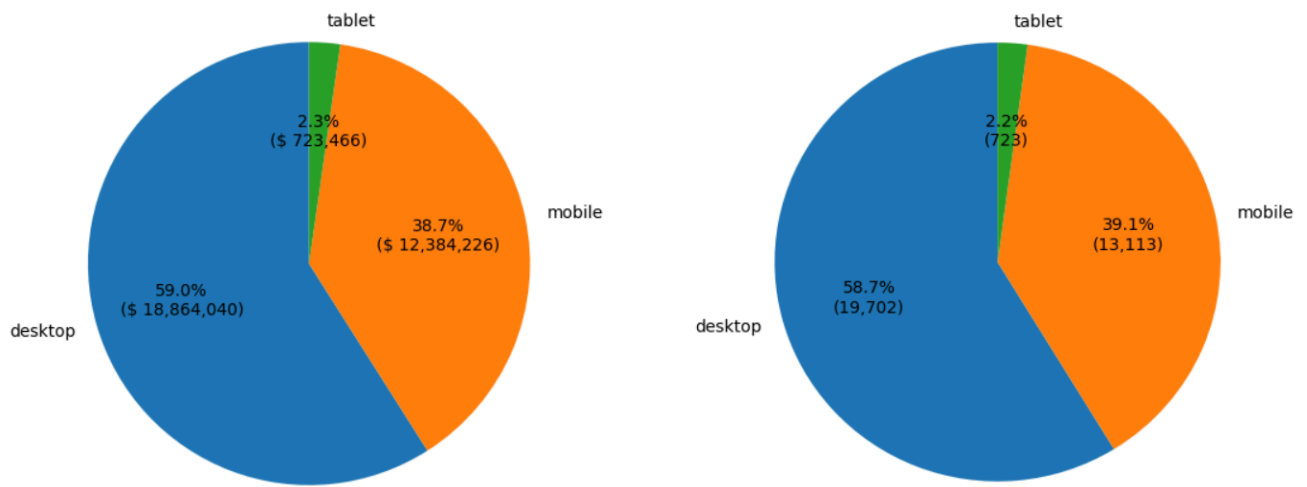


Figure 8. Revenue by Device Type Figure 9. Number of Orders by Device Type

Sales performance by traffic channel is presented in Figures 10 and 11.

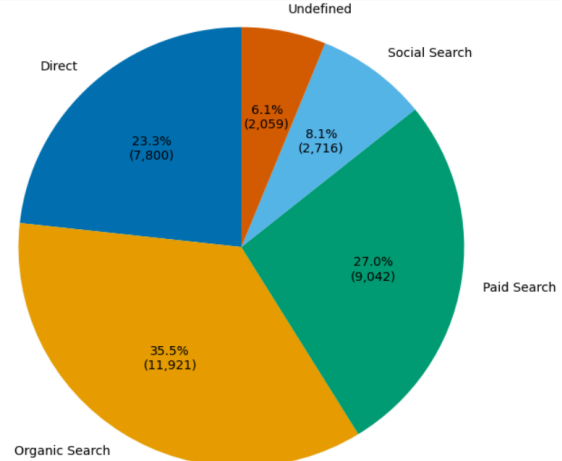
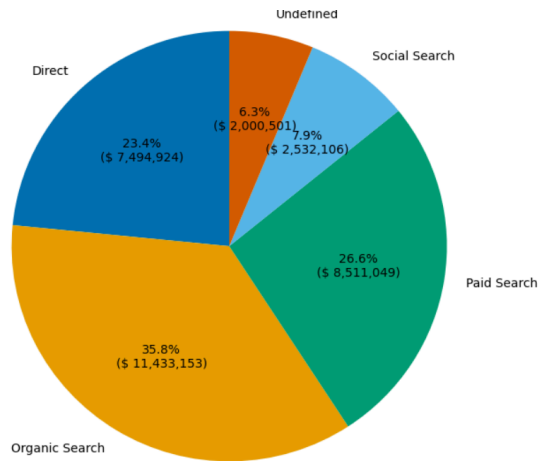


Figure 10. Revenue Distribution by Traffic Channel Figure 11. Number of Orders by Device Type

The total number of registered users during the reporting period was 27,945, of which 20,036 (71.7%) confirmed their email address, and 4,735 (16.9%) unsubscribed from the mailing list.

The average order value among users who unsubscribed was \$965.8, while for those who remained subscribed, it was \$921.5. This indicates that opting out of company communications does not reflect a decline in interest in the products.

The majority of registered users reside in the United States. Overall, users are registered in 108 countries. Details on user counts in the top 10 countries and by continent are provided in Tables 4 and 5.

	country	number of registered users
1	United States	12,384
2	India	2,687
3	Canada	2,067
4	United Kingdom	859
5	France	553
6	Spain	536
7	Taiwan	500
8	China	490
9	Germany	490
10	Italy	386

Table 4. Top 10 Countries by Number of Registered Users

	continent	number of registered users
1	Americas	15,520
2	Asia	6,664
3	Europe	5,170
4	Oceania	272
5	Africa	266
6	(not set)	53

Table 5. Distribution of Registered Users by Continent

The dynamics of total session counts, as well as sessions involving user registrations and purchases, are shown in Figure 12. Their distribution by traffic channels and devices is presented in Tables 6 and 7, while revenue by channels and devices is detailed in Table 8.

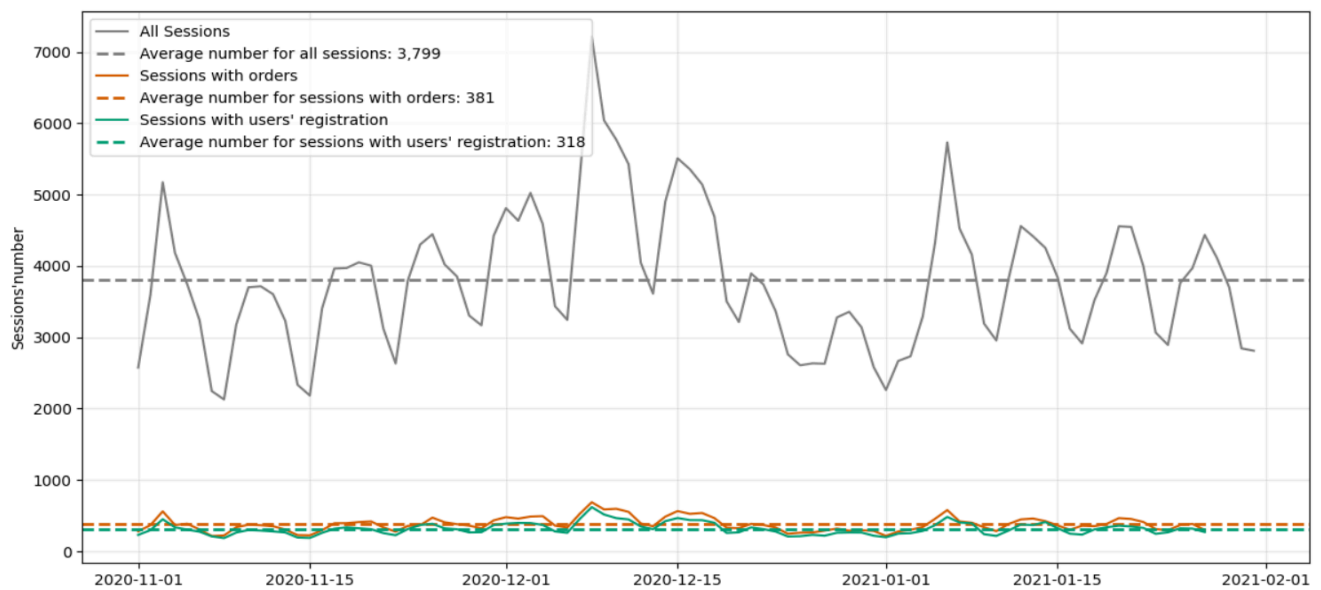


Figure 12. Session Dynamics

channel	number_of_sessions	sessions_with_orders	sessions_with_registrations
Organic Search	124425	11921	10035
Paid Search	94341	9042	7467
Direct	81382	7800	6514
Social Search	27914	2716	2212
Undefined	21483	2059	1717

Table 6. Total Number of Sessions by Traffic Channel

device	number_of_sessions	sessions_with_orders	sessions_with_registrations
desktop	204429	19702	16417
mobile	137267	13113	10878
tablet	7849	723	650

Table 7. Total Number of Sessions by Device Type

device	desktop	mobile	tablet	Total
channel				
Direct	4655	2985	160	7800
Organic Search	7011	4655	255	11921
Paid Search	5261	3574	207	9042
Social Search	1594	1065	57	2716
Undefined	1181	834	44	2059
Total	19702	13113	723	33538

Table 7. Number of Sessions with Orders by Traffic Channel and Device Type

Sales Dynamics Analysis Figures 13–15 illustrate revenue trends overall and broken down by continent, traffic channel, and device type.

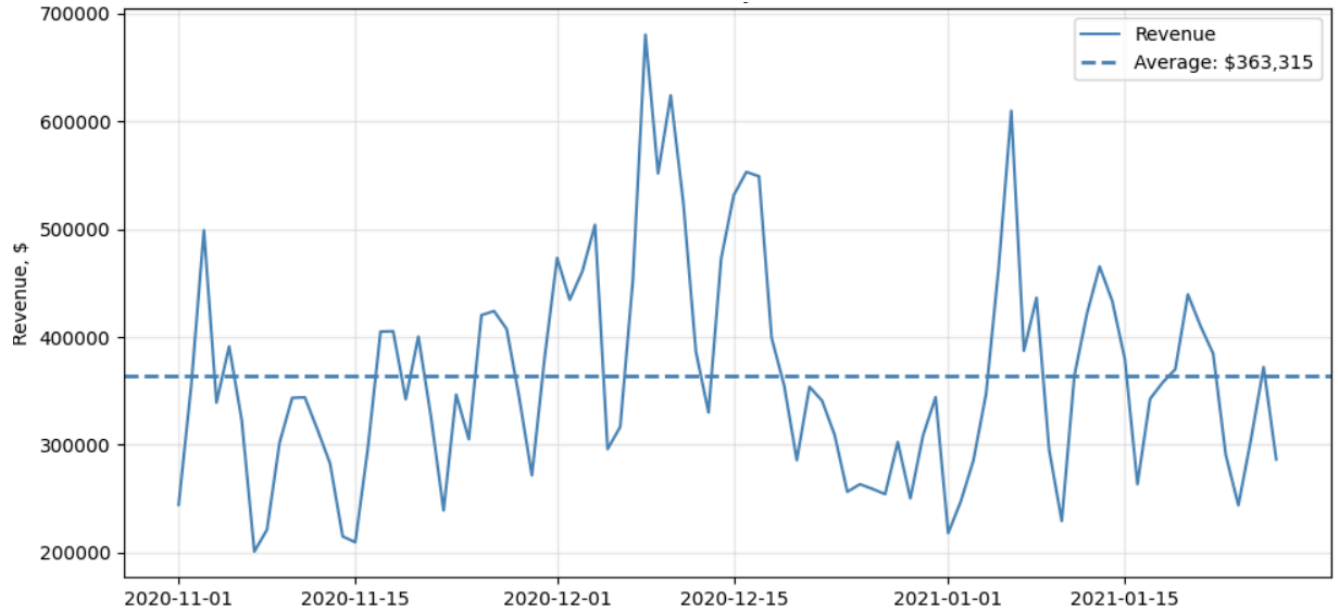


Figure 13. Daily Revenue Dynamics

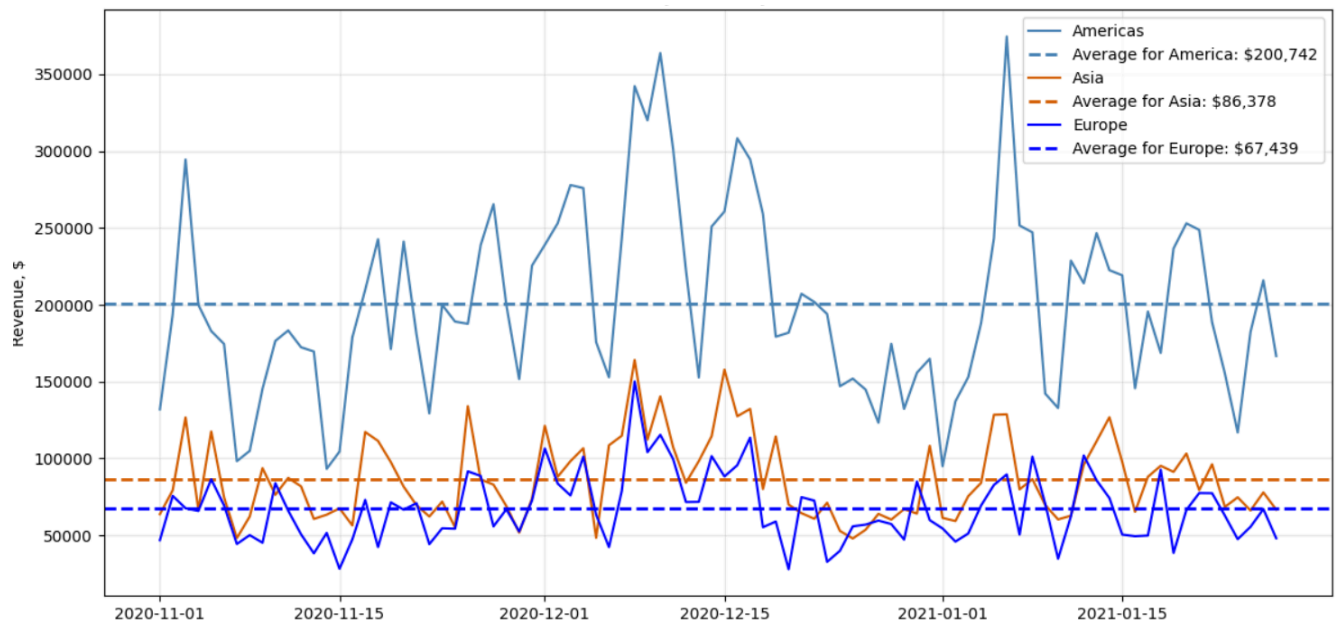


Figure 14. Revenue Dynamics by Continent

Correlation and p-value analysis indicate that sales across the top three continents are strongly correlated:

- Americas vs Europe: $r = 0.670$, $p\text{-value} = 0.00000$

- Americas vs Asia: $r = 0.692$, $p\text{-value} = 0.00000$
- Europe vs Asia: $r = 0.668$, $p\text{-value} = 0.00000$

Correlation coefficients ranging from 0.66 to 0.69 reflect a moderately strong positive relationship, meaning that when sales increase on one continent, similar growth tends to occur on the others. The extremely low p-values ($\ll 0.05$) confirm that these correlations are statistically significant, suggesting the relationships are unlikely to be random.

This pattern may indicate global sales trends, such as peak days or seasonal effects that simultaneously influence all continents.

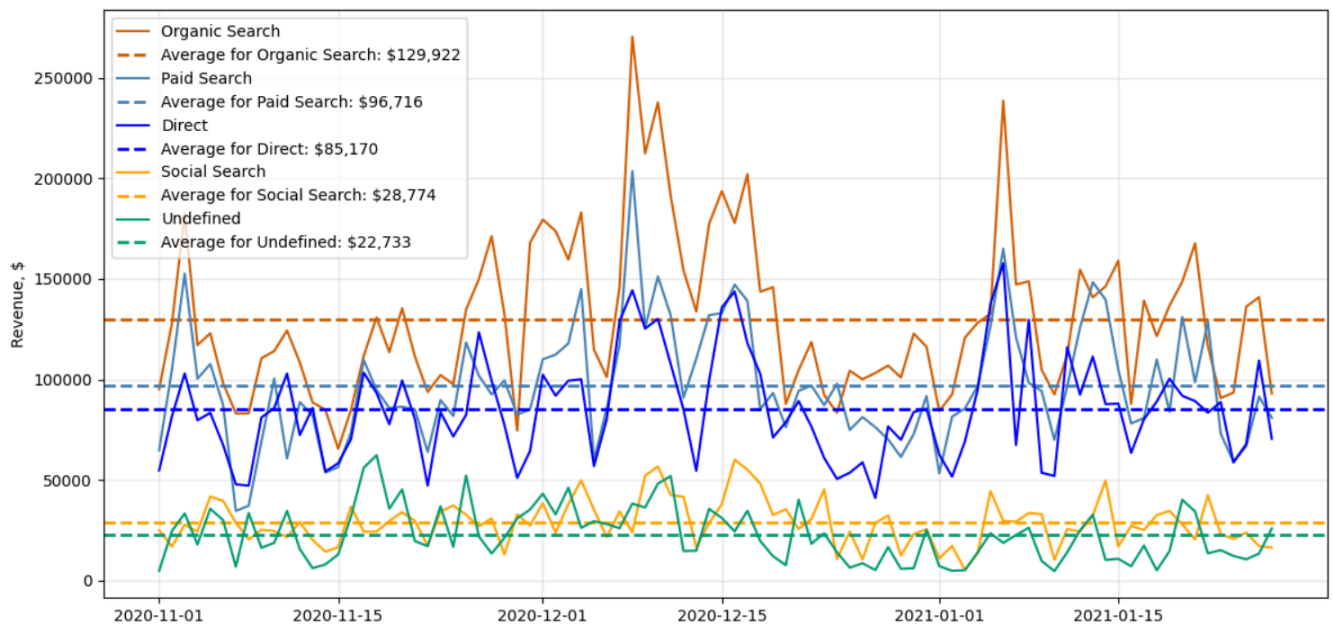


Figure 15. Revenue Dynamics by Traffic Channel

The results of the correlation analysis reveal a clear positive relationship between sales across different traffic channels.

Пара каналів	r (коефіцієнт)	$p\text{-value}$	Висновок
Organic Search – Paid Search	0.808	0.00000	Сильна, статистично значуща кореляція
Organic Search – Direct	0.760	0.00000	Сильна, значуща

Organic Search – Social Search	0.434	0.00002	Помірна, значуща
Paid Search – Direct	0.724	0.00000	Сильна, значуща
Paid Search – Social Search	0.452	0.00001	Помірна, значуща
Direct – Social Search	0.462	0.00001	Помірна, значуща

All p-values are below 0.05, indicating that all identified correlations are statistically significant.

The highest Pearson correlation coefficients ($r \approx 0.8$, 0.76, 0.72) are observed between Organic Search, Paid Search, and Direct channels. This suggests that sales in these channels tend to fluctuate together — likely due to shared influences such as overarching marketing campaigns or seasonal peaks.

Social Search shows a moderate correlation ($r \approx 0.43$ –0.46), indicating a somewhat distinct sales dynamic. This may imply that the impact of social media does not fully align with the trends observed in other channels.

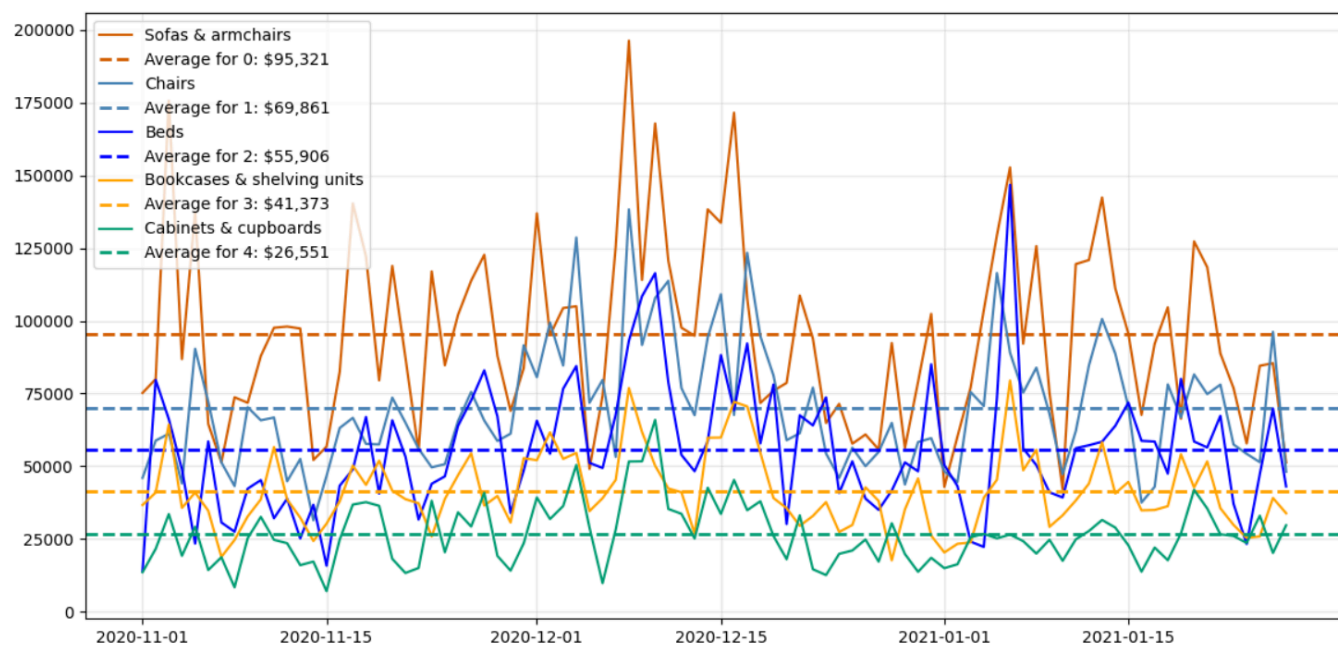


Figure 16. Revenue Dynamics by Product Category

Correlation and p-value analysis between product categories:

- Sofas & armchairs vs Chairs: $r = 0.578$, $p\text{-value} = 0.00000$
- Sofas & armchairs vs Beds: $r = 0.535$, $p\text{-value} = 0.00000$
- Sofas & armchairs vs Bookcases & shelving units: $r = 0.666$, $p\text{-value} = 0.00000$
- Sofas & armchairs vs Cabinets & cupboards: $r = 0.658$, $p\text{-value} = 0.00000$
- Chairs vs Beds: $r = 0.554$, $p\text{-value} = 0.00000$
- Chairs vs Bookcases & shelving units: $r = 0.637$, $p\text{-value} = 0.00000$
- Chairs vs Cabinets & cupboards: $r = 0.573$, $p\text{-value} = 0.00000$
- Beds vs Bookcases & shelving units: $r = 0.593$, $p\text{-value} = 0.00000$
- Beds vs Cabinets & cupboards: $r = 0.515$, $p\text{-value} = 0.00000$
- Bookcases & shelving units vs Cabinets & cupboards: $r = 0.539$, $p\text{-value} = 0.00000$

All correlation coefficients (r) fall within the range of 0.5–0.67, indicating moderate to strong positive correlations between product categories. All p -values are 0.00000, meaning all correlations are statistically significant ($p < 0.05$). This suggests that sales dynamics in one product group often mirror those in others — increases or decreases in one category tend to coincide with similar trends in others.

Seasonal Patterns The revenue dynamics charts above reveal clear seasonal patterns, likely reflecting weekly seasonality, where customer activity increases on specific days of the week. Additionally, there are several notable spikes in customer activity in mid-December and early January, likely linked to promotional events or seasonal sales.

These observations are further supported by the upcoming charts, which illustrate sales patterns by day of the week and by month (Figures 17 and 18).

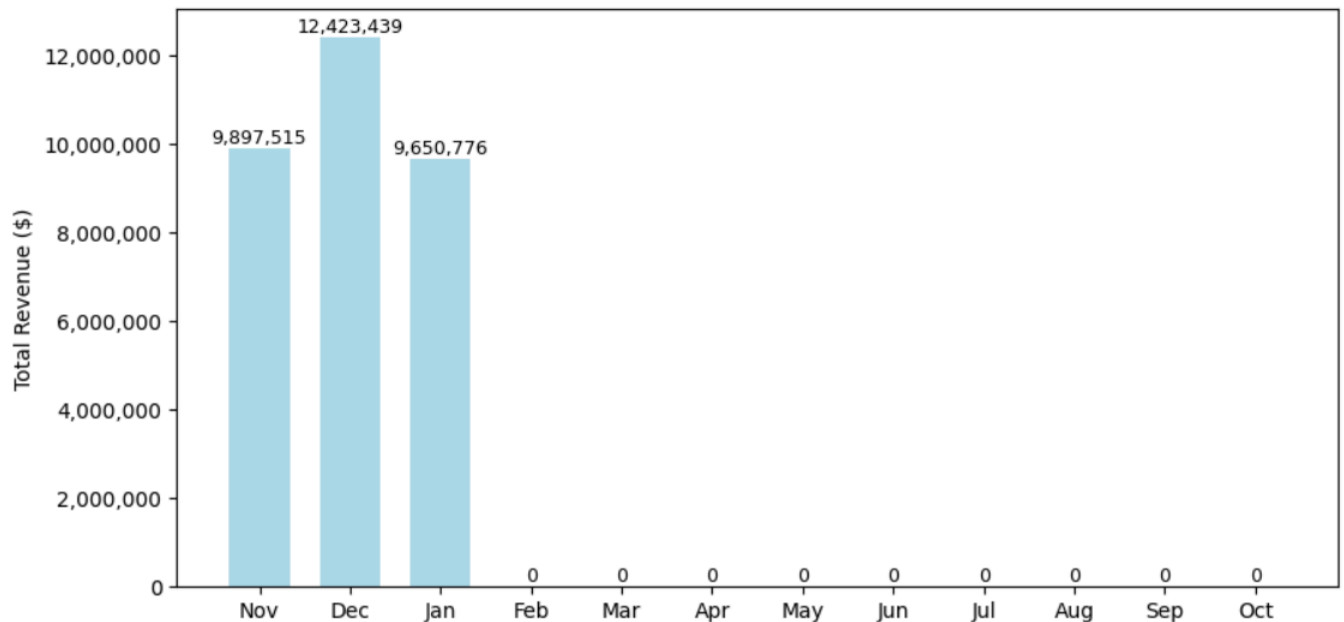


Figure 17. Total Revenue by Month

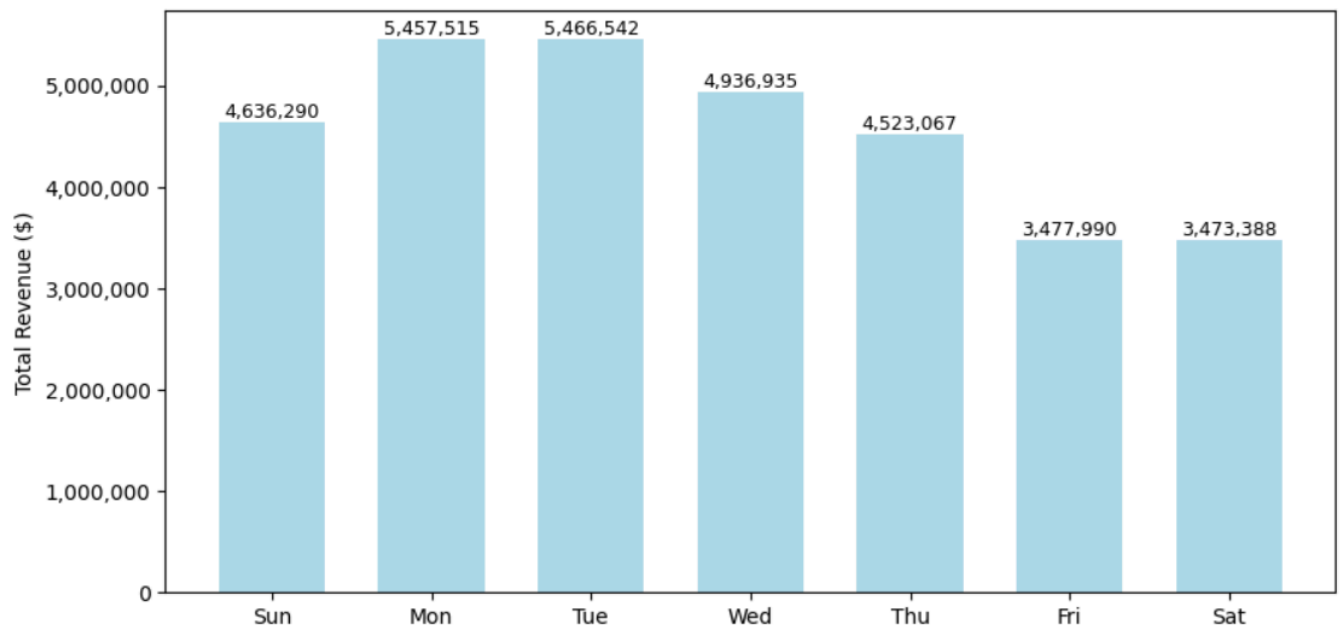


Figure 18. Total Revenue by Day of the Week

A strong correlation between revenue level and daily session count is shown in Figure 19.

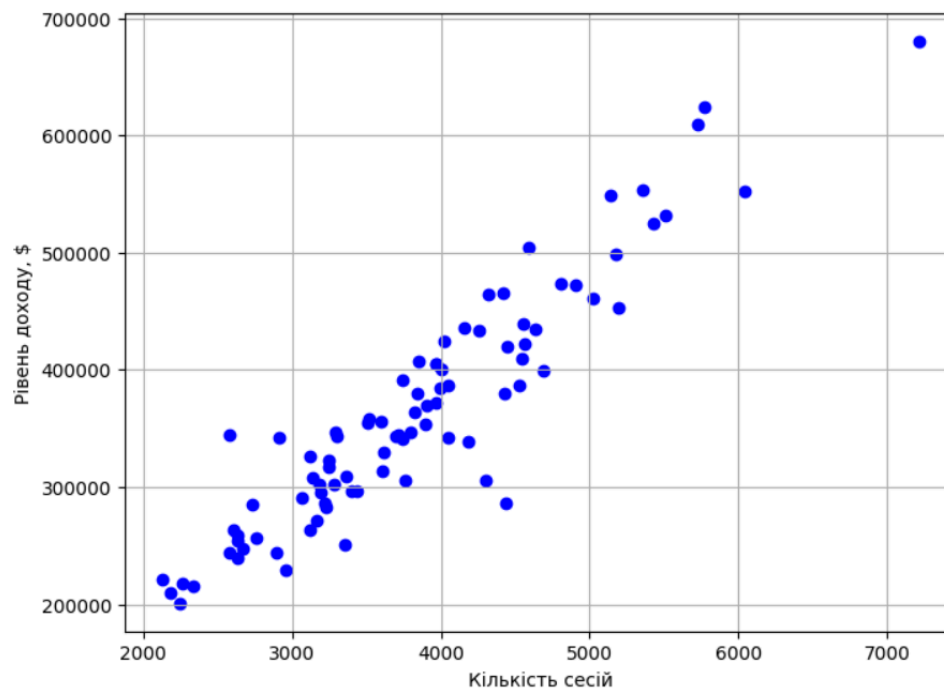


Figure 19. Correlation Between Daily Session Count and Revenue

There is a very strong positive correlation between these indicators. This is confirmed by the following statistical coefficients: $p\text{-value} = \text{np.float64}(1.7394621412998283\text{e-}39)$ — an extremely low p-value, indicating statistical significance of the correlation. In other words, the probability that this correlation is random is virtually zero. Pearson Correlation = 0.9313146607045042 — very close to 1, meaning that the more sessions there are, the more sales occur.

The level of sales and the number of sessions per day are strongly connected, and this relationship is highly reliable. Sales dynamics by registered and unregistered users are shown in Figure 20.

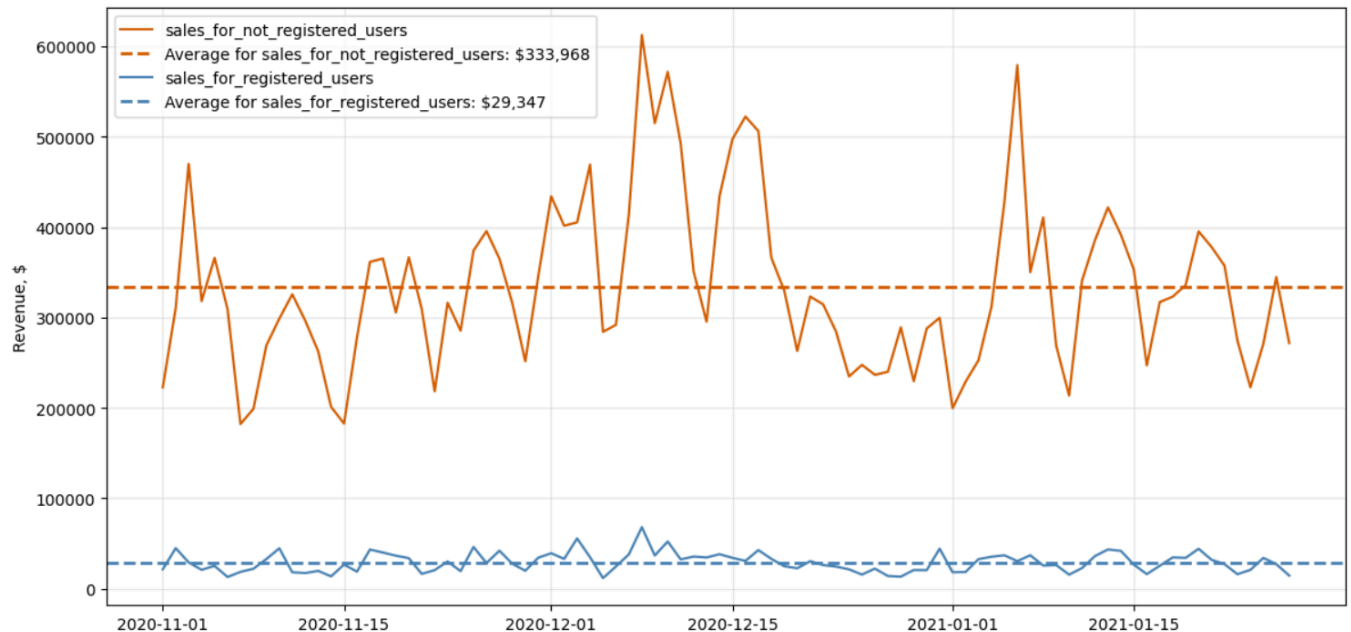
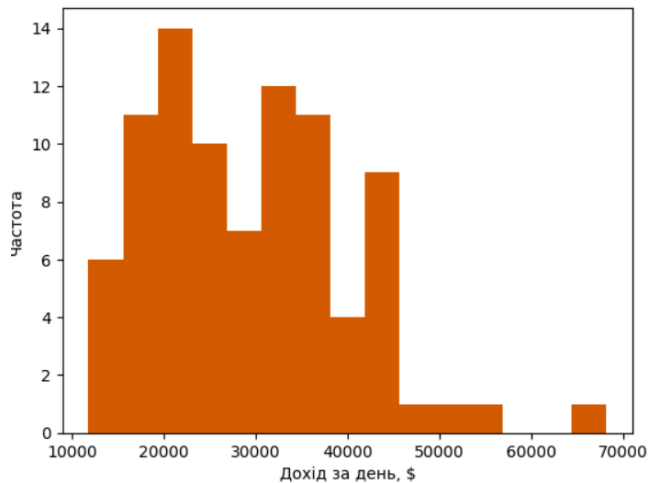
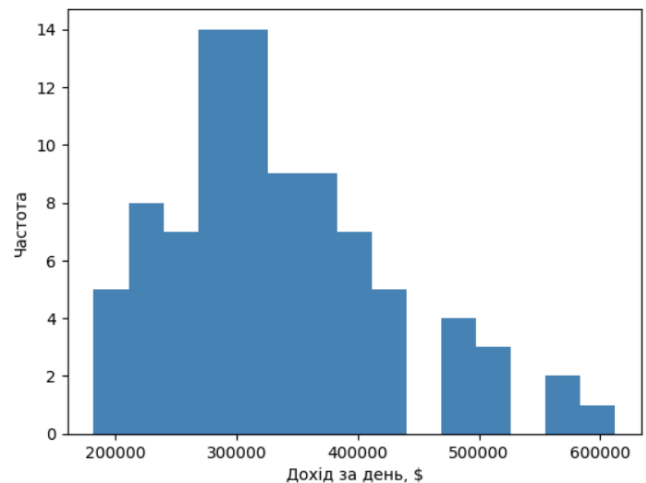


Figure 20. Sales Dynamics by Registered and Unregistered Users

To analyze the differences between these groups, visualizations of their value distributions were created (Figures 21, 22).



Малюнок 21. Розподіл щоденних продажів зареєстрованим користувачам



Малюнок 22. Розподіл щоденних продажів незареєстрованим користувачам

Враховуючи розмір вибірок, проведено тест Д'агостіно-Пірсона на визначення ступеню нормальності розподілу. Результати тесту свідчать що розподіл цих двох наборів даних суттєво відрізняється від нормального (для зареєстрованих користувачів $p\text{-value} = 0.01$, для незареєстрованих користувачів $p\text{-value} = 0.006$).

Оскільки обидві групи не є нормально розподіленими, для їх порівняння використано непараметричний тест Манна-Уїтні. Отриманий коефіцієнт $p\text{-value} = 0.00$ означає, що різниця між вибірками є статистично значущою.

Коефіцієнт кореляції Пірсона ($r = 0.659$) підтверджує досить сильний додатний зв'язок між продажами для зареєстрованих і незареєстрованих користувачів (Малюнок 23). Тобто, коли зростають продажі серед зареєстрованих, зазвичай зростають і серед незареєстрованих. Коефіцієнт $p\text{-value} = 0.00000$ показує, що результат статистично значущий, тобто визначений додатний зв'язок є не випадковим.

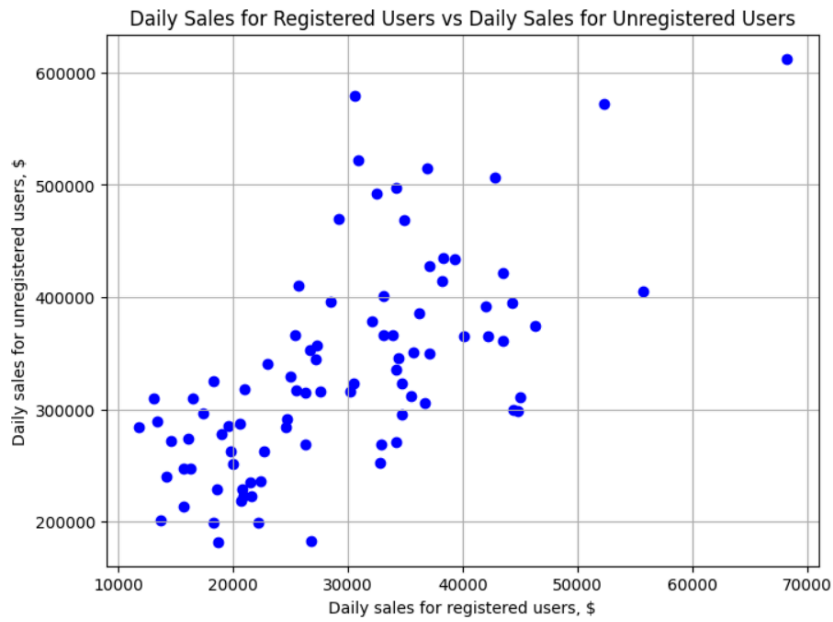


Figure 23. Daily Sales to Registered Users vs Daily Sales to Unregistered Users

The dynamics of daily session counts across traffic channels are shown in Figure 24.

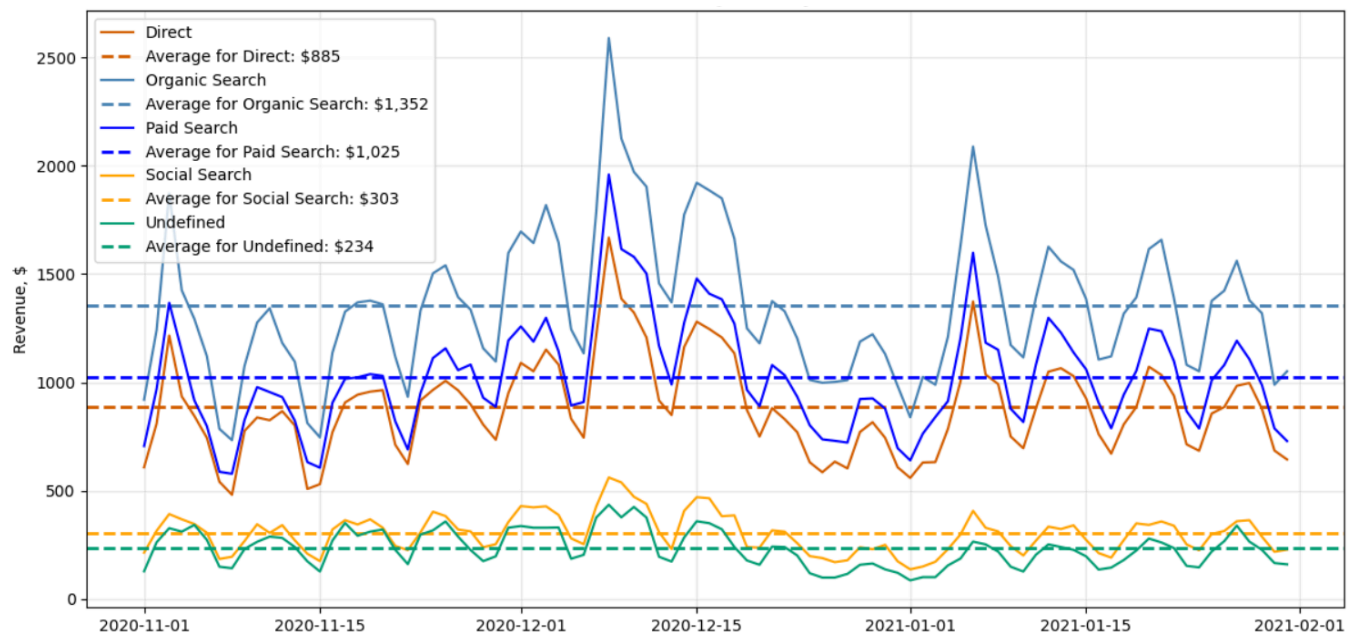


Figure 24. Daily Session Count Dynamics Across Traffic Channels

Regarding the number of sessions across different traffic channels, the following can be noted: The distribution of daily session counts in the Social Search channel is normal, as confirmed by the D'Agostino–Pearson test ($p\text{-value} = 0.2144$) and illustrated in the visualization (Figure 25).

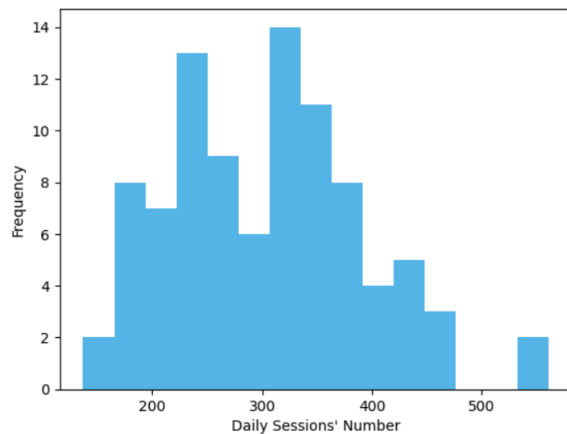
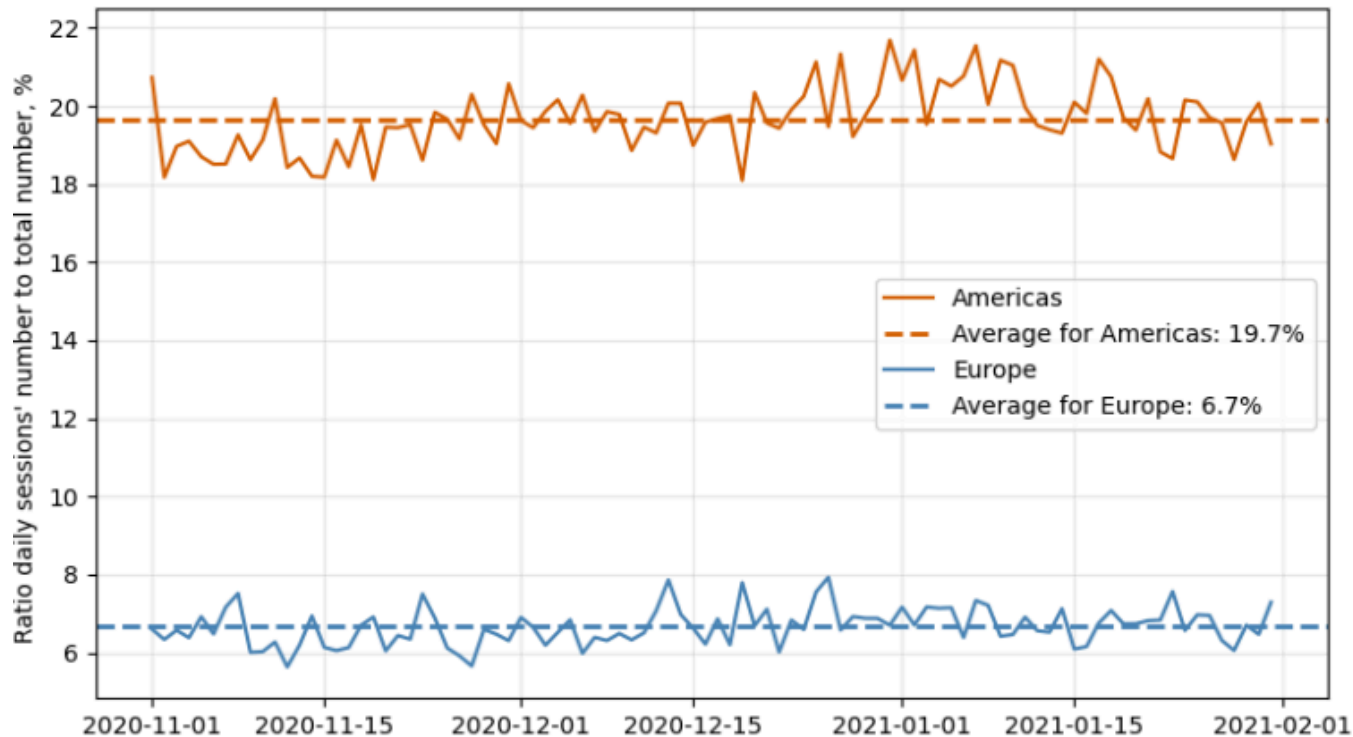


Figure 25. Distribution of Daily Session Count via the Social Search Channel

The distribution of session counts across other channels does not follow a normal distribution ($p\text{-value} < 0.05$). Given this, the Kruskal–Wallis test was used to analyze relationships between datasets. The results showed statistically significant differences between groups ($p\text{-value} = 0.00$). For pairwise group comparisons, the Mann–Whitney U test was applied. The obtained coefficients ($p\text{-value} = 0.00$) indicate statistically significant differences between all groups. To assess the degree of correlation between groups, Spearman correlation coefficients were calculated, as normality was not observed (based on the normality test). All correlation coefficients were $r > 0.78$, with most exceeding 0.9, indicating a very strong positive relationship between session dynamics across different channels — meaning that when session counts increase in one channel, they almost always increase in others. All $p\text{-values}$ were significantly below 0.05, confirming that all correlations are statistically significant. Session count dynamics across all channels show a strong and statistically significant positive relationship. This suggests that fluctuations in user activity across channels occur synchronously — likely influenced by common factors such as seasonality, marketing campaigns, or news events.

The dynamics of the daily share of sessions from organic traffic in Europe and the Americas are shown in Figure 26.



Малюнок 26. Динаміка щоденної долі сесій з органічним трафіком у Європі та Америці

Для перевірки наборів даних на відповідність нормальному розподілу використано тест Д'агостіно-Пірсона, результати якого показали, що ці набори даних можна вважати нормальними ($p\text{-value}=0.506$ - Америка, $p\text{-value}=0.278$ - Європа).

Враховуючи це, для визначення ступеню кореляції застосовано тест Пірсона, результати якого свідчать про дуже слабкий позитивний зв'язок між змінними ($r=0.088$) при цьому зв'язок не є статистично значущим ($p\text{-value}=0.404$).

Відсутність взаємозв'язку між цими наборами даних можна побачити на малюнку 27.

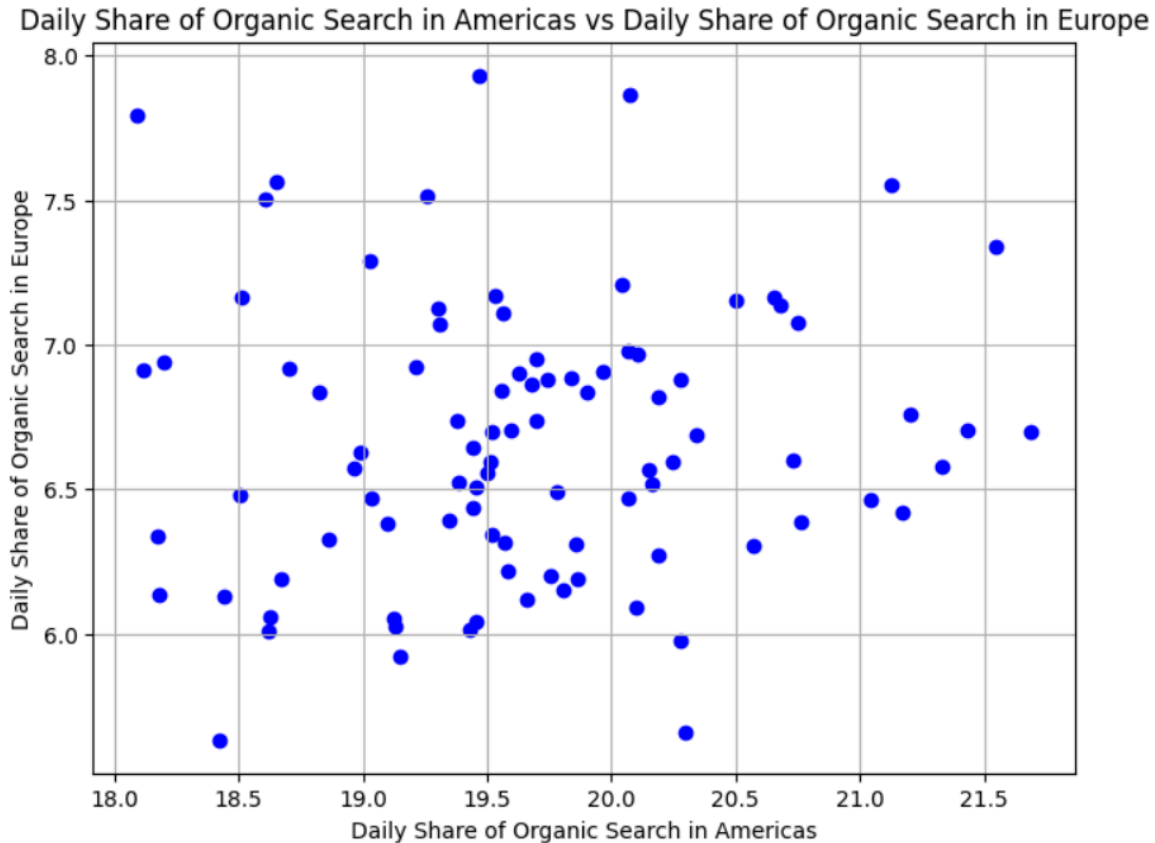


Figure 27. Daily Share of Organic Search in the Americas vs Daily Share of Organic Search in Europe

As a result of statistical analysis of sessions and sales dynamics across continents, traffic channels, and product categories, the following findings were established:

Distribution Normality

For most samples, the distribution of indicators did not follow a normal pattern (based on the D'Agostino–Pearson test, $p < 0.05$), which led to the use of non-parametric methods (specifically, the Kruskal–Wallis test) for group comparisons.

In cases where the samples had a normal distribution, parametric analysis was applied (Pearson's test, t-test).

Continent Comparison

Sales across the top three continents (Americas, Europe, Asia) showed a moderately strong positive correlation ($r \approx 0.67$ – 0.69 , $p < 0.001$), indicating consistent demand trends across different markets.

Traffic Channel Comparison

Among the main channels (Organic, Paid, Direct, Social), a strong positive correlation was observed ($r \approx 0.7$ – 0.8 , $p < 0.001$), meaning increased activity in one channel was accompanied by growth in others. This may indicate synergy between marketing channels.

Product Category Comparison

Sales across major product categories also showed a moderate positive correlation ($r \approx 0.5\text{--}0.6$, $p < 0.001$), suggesting shared seasonality or behavioral patterns among consumers.

Registered vs Unregistered Users

A significant positive correlation was found between sales in both groups ($r \approx 0.66$, $p < 0.001$), which may reflect general demand trends regardless of user registration status.

Organic Search: Americas vs Europe

Pearson's test ($r = 0.088$, $p = 0.404$) showed no statistically significant relationship between the indicators of the two regions — their dynamics are likely driven by different market conditions.

General Conclusion

The analysis revealed stable positive interconnections across most channels, categories, and continents, indicating an interconnected structure of sales and traffic.

At the same time, certain markets (particularly the Americas and Europe in the Organic Search channel) demonstrate independent behavior, which may reflect distinct marketing or consumer characteristics.

These findings should be considered when planning campaigns targeted at specific regions or user segments.